

Reservoir Stage Record - Numerical Methods Laboratory Dashboard

Laboratory Activity - Reservoir Stage Analysis

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Generated 2026-08-27 22:39:36 | Data: Sensor Log (15-min logger) | N = 288 pts, Δt = 0.25 h

FITTED PARAMETERS — SUM OF TWO LOGISTICS MODEL

$$\hat{h}(t) = c + a_1/(1+e^{-(k_1(t-t_{01}))}) + a_2/(1+e^{-(k_2(t-t_{02}))})$$

Parameter	Estimate	SE	t	p (2-tailed)
a ₁ (primary amplitude, m)	8.915647	0.75310	11.84	< 0.001
k ₁ (primary rate, 1/h)	0.511739	0.02128	24.05	< 0.001
t ₀₁ (primary inflection, h)	31.519791	0.09055	348.10	< 0.001
a ₂ (secondary amplitude, m)	-3.720879	0.75936	-4.90	< 0.001
k ₂ (secondary rate, 1/h)	0.283130	0.01375	20.59	< 0.001
t ₀₂ (secondary inflection, h)	37.546688	1.05747	35.51	< 0.001
c (baseline, m)	14.349259	0.00968	1483.06	< 0.001

All parameters highly significant (two-tailed t-tests, dof = 281).

GOODNESS-OF-FIT STATISTICS

SSE	1.8017
R ²	0.999114
n (obs.)	288

SSR	2032.2764
R ² (adj.)	0.999095
p (params)	7

SST	2034.0779
s (m)	0.0801
dof (n-p)	281

AREA UNDER FITTED CURVE — NUMERICAL INTEGRATION

Method: `scipy.integrate.quad` on the continuous fitted model $\hat{h}(t)$; not on the raw discrete points.
Integration bounds: t = 0.00 to 71.75 h (duration 71.75 h)

Abs. error (quad)	5.1e-07	Trap. (raw pts)	1260.9963 m·h	Diff. (quad - trap)	-0.025 m·h	Mean fitted stage	17.5745 m
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A = ∫ $\hat{h}(t)$ dt = 1260.9708 m·h
(meter-hours — time integral of reservoir depth above gauge datum)

READING OF THE RESIDUALS

Residual mean	2.28e-09 m
Durbin-Watson	0.066

Residual SD	0.0792 m
Sign changes	16 / 288

Max residual	0.221 m
Positive / Negative	134 / 154

- Centred: Residuals are centred on zero (mean = 2.28e-09 m).
- Runs: Long runs of the same sign remain (only 16 sign changes in 288 points; 134 positive, 154 negative residuals), indicating mild systematic structure.
- Spread: Residual spread does not widen appreciably with fitted level (homoscedastic appearance).
- Largest: Largest |residual| = 0.221 m at t = 30.25 h (2026-07-22 06:15), which is >20× the 0.01 m logger resolution.
- Autocorrelation: Durbin-Watson statistic ≈ 0.066 (values near 2 suggest little autocorrelation; lower values indicate positive serial correlation).